

## Introduction

The last decade has witnessed tremendous controversy surrounding the ethics of clinical research. There have been disagreements over the use of placebos in developing countries when there are effective but costly therapies available in more developed countries; there has been uncertainty over researchers' obligations to participants once trials are complete; there have been disagreements over research with children and mentally incapacitated patients; and there is widespread condemnation of academic researchers' financial conflicts of interest. The deaths of Jesse Gelsinger and Ellen Roche while both were enrolled in clinical research, and the suspension of clinical research at Duke, Johns Hopkins, and other major research institutions has created concern over the safety of clinical research. Suppression of data on adverse drug events by pharmaceutical corporations has created concern over the integrity of both researchers and the research enterprise. Probably nothing signifies this controversy and its public nature better than the April 22, 2002, issue of *Time* magazine, whose cover depicted a human being in a hospital gown inside a cage, under a caption that read, "How Medical Testing Has Turned Millions of Us into . . . Human Guinea Pigs."<sup>1</sup>

Of course, this is not the first era of controversy surrounding clinical research. At least three periods of sustained controversy have occurred before this latest decade. In the late 19th century, there was an important controversy surrounding the search for the cause and a cure of yellow fever. As Susan Lederer describes in Chapter 1, Guiseppe Sanarelli was an Italian researcher working in South America. Working with the legacy of Koch and Pasteur's isolation of microorganisms as causes of disease, he declared that he had isolated the bacillus that caused yellow fever, and was able to induce yellow fever in five people by infecting them with the

agent. (Leave aside that yellow fever is caused not by a bacillus but by a virus.) His work was categorically condemned by many researchers, most importantly by William Osler, at the time the world's most prominent and famed physician and chairman of medicine at Johns Hopkins Medical School and soon to be Regius Professor of Medicine at Oxford. At a professional meeting in 1898, Osler declared, "To deliberately inject a poison of known high degree of virulency into a human being, unless you obtain that man's sanction, is not ridiculous, it is criminal."<sup>2</sup>

After the yellow fever controversy there was the entire episode of Nazi medicine and medical research. As delineated by Paul Weindling in Chapter 2, this period entailed a myriad of the most gruesome and horrific experiments, from placing people in freezing water until they died to subjecting them to very low atmospheric pressures until they exploded, from injecting them with typhoid to Mengele's twin experiments. Part of the horror was how many German researchers were able to use their research samples and data after the war to continue as respected researchers.

Then beginning in the early 1960s, there was a series of research scandals in the United States that culminated in the revelations about the Tuskegee Syphilis Study. This period began in July 1963 with the Brooklyn Jewish Chronic Disease Hospital case, which is described by John Arras in Chapter 6. Prominent cancer researchers "injected live cancer cells into 22 chronically ill and debilitated patients" without informing them that cancer cells were being used. Patients were unaware that this was not a therapeutic intervention to treat their condition, but was rather an experiment to gain scientific knowledge. Furthermore, the patients were not asked for their consent to the study.

Then in 1966, Henry K. Beecher, a prominent anesthesiologist at the Massachusetts General Hospital and professor at Harvard Medical School, published a paper in the *New England Journal of Medicine* entitled “Ethics and Clinical Research.”<sup>3</sup> In it, he delineated 22 cases that he claimed were extracted from an original list of 50 cases. He noted that these cases came “from leading medical school, university hospitals, private hospitals, governmental military departments (the Army, the Navy, and the Air Force), governmental institutes (the NIH), Veterans Administration hospitals, and industry.” Beecher wrote that these cases represented “troubling practices” in which many of the patients never had the risk satisfactorily explained to them, and further hundreds did not know that they were the subjects of an experiment although they suffered grave consequences as a direct result.

It should be noted that Beecher may not have been careful in all of the 22 cases he criticized. As Walter Robinson and Brandon Unruh note in Chapter 7, although Beecher strongly condemned the Willowbrook hepatitis studies—which left an enduring taint upon its principal investigator, Dr. Saul Krugman—this research seems to have fulfilled ethical requirements and, upon closer examination, to have been conducted in an ethical manner.

As bad as some of Beecher’s cases were, worse was revealed in 1972 when the Tuskegee Syphilis Study was disclosed to the public. As described by James Jones in Chapter 8, this study has a great many troubling and unethical aspects. Not only was the scientific justification of the study questionable when it was initiated in the 1930s, the actual trial entailed multiple layers of deception, and trial personnel actively prevented the participants from getting medication—penicillin—to which they were entitled. In the 1960s, a social worker working for the U.S. Public Health Service in San Francisco, Peter Buxton, learned about Tuskegee from coworkers and launched a one man crusade to stop it. As Jones notes, this prompted an internal ethical evaluation of the study, which ultimately sanctioned continuing the study. Only when subjected to public scrutiny through an Associated Press story in 1972 did the trial get halted by the Secretary of the Department of Health Education and Welfare.

Of course, these are not the only scandals or controversies involving clinical research. There were less well known but nonetheless strong condemnations of the appalling research experiments committed by the Japanese military in World War II, well described by Takashi Tsuchiya in Chapter 3. In the 1980s, as John Killen notes in Chapter 9, debate surrounded trials of drugs for HIV/AIDS and whether the research subject protections were excessively paternalistic. Importantly, a big change occurred in this era when research participants were major participants in the debates and, in a surprise to many, challenged the oversight of clinical research as excessively protectionist and paternalistic.

In one sense, such scandals and debates have been quite beneficial. These controversies have forced the reexamination of fundamental issues long deemed settled and produced much of the ethical guidance for clinical research. As Carol Levine has observed, our approach to the ethics of clinical research was “born in scandal and reared in protectionism.”<sup>4</sup> Although this may overstate the case a bit, it is certainly true that the condemnation of Sanarelli’s claims led Walter Reed to carefully reflect on his yellow fever studies and to delineate five key safeguards: (1) auto-experimentation, with the researchers serving as participants (including the fact that one, Jesse Lazear died as a result); (2) use only of adult participants; (3) signed, written contracts with research

participants; (4) financial payment to the participants; and (5) declaration in the papers that each participant gave his consent, forerunner to the current practice of indicating in published papers that the research was conducted in accord with ethical guidelines. The judicial decision in the post war trial of the Nazi doctors, *United States v. Karl Brandt et al.*, articulated the Nuremberg Code.<sup>5</sup>

In the wake of the Jewish Chronic Disease Hospital case and other scandals, the U.S. Public Health Service required independent review of research studies to assess the risk-benefit ratio and the adequacy of measures to obtain informed consent. The Tuskegee scandal prompted the creation of the National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research by the U.S. Congress. In April 1979, this Commission issued its Belmont Report, which articulated respect for persons, beneficence, and justice as the “broader ethical principles [to] provide a basis on which specific rules may be formulated, criticized, and interpreted.”<sup>6</sup> The Commission also induced the Department of Health, Education and Welfare to adopt regulations requiring that institutional review boards review all research protocols before they were begun and other safeguards. Similarly, worries about HIV studies in developing countries led to the International Ethical Guidelines for Biomedical Research Involving Human Subjects promulgated by the Council for International Organizations of Medical Sciences (CIOMS).<sup>7</sup>

Although forcing needed change, these controversies have also stymied clinical research. They have made well-intentioned researchers uncertain of how they can act ethically and how to design an ethical clinical research trial. For instance, ongoing disagreement about the ethics of using placebo controls in developing countries that cannot afford expensive medications or interventions may have inhibited the initiations of studies in developing countries; disagreement about researchers’ obligations to health needs of participants that are unrelated to the purposes of research has prevented studies from occurring; different views on what benefits must be provided at the conclusion of trials have generated charges of exploitation; and controversy about the ethics of paying research participants has led many to condemn studies because they are “coercive.”

One problem with these controversies is that although bioethicists, research ethics boards, regulatory agencies, international organizations, and others debate what is the ethical thing to do, research must proceed and researchers must make decisions in planning and conducting research studies. This creates challenges for researchers and induces worries that even actions chosen conscientiously and with good intentions may be charged with being unethical. This may cast a chill on research.

Another problem is that the disputes have generated huge amounts of literature—commentaries, conceptual analyses of issues, and empirical research studies. The growth of the literature is not a problem per se. Indeed, good analyses can lead to wider and deeper understanding of many issues. But poor analyses can inappropriately condemn actions and studies as “exploitative,” “coercive,” or “unjust,” thereby inhibiting perfectly ethical research. In addition, a vast literature may make it hard for experts in research ethics, let alone clinical researchers, to be able to know what is the current status of an issue, such as payment to research participants, use of stored biological samples, improving informed consent, or enrollment of people who are mentally incompetent, or what constitutes minimal risk. Although there have been

comprehensive review articles on a few relevant topics, there has not been a comprehensive and systematic synthesis and critical analysis of the research ethics literature in all areas. The need for comprehensive reviews is especially critical when it comes to synthesizing empirical studies. Such reviews require collecting large numbers of studies, synthesizing data based on different methodologies, and critically comparing results. The reviews that do exist have generated results that have challenged common assumptions and surprised many people, including experts in the field. For instance, the common assumption that using videos or interactive computer programs will enhance understanding during the informed consent process has been challenged by a comprehensive review. Similarly, the contention that the quality of informed consent is worse in developing countries compared to developed countries was not substantiated when all available studies were analyzed.

A third problem relates to the available ethical guidance. Many codes or declarations were responses to scandals. Walter Reed's five principles were a response to Sanarelli's ethical violations; the Nuremberg Code was a response to the Nazi war crimes; the Belmont Report was a response to Tuskegee Syphilis Study; the ethical guidelines of the Advisory Committee on Human Radiation Experiments was a response to the radiation experiments in the United States; and the latest revision of the Declaration of Helsinki<sup>8</sup> was a response to the controversy surrounding placebo-controlled trials in developing countries. A major problem with such response to scandals is that these guidelines, codes, and declarations focus on the specific issue or issues raised by the scandal and use the scandal as the litmus test for their recommendations. There has been little effort to reflect on a more general ethical framework to guide research ethics, free of the emotional outrage based on the latest egregious violations when, as Bishop Butler put it, "we sit down in a cool hour."

One consequence is that there are contradictions among these various "authoritative" guidelines. For instance, the Nuremberg Code does not permit research with people who cannot consent—children and mentally incapacitated individuals—whereas other guidelines permit such research under certain circumstances; the Declaration of Helsinki seems to prohibit placebo controls whenever there is a proven treatment, whereas most other guidance, including that from CIOMS, the Nuffield Council,<sup>9</sup> and the National Bioethics Advisory Commission,<sup>10</sup> permits such trials under certain circumstances; and CIOMS requires reasonable availability of a drug in a country if it has been proven effective in that country, but this is not required by other guidance including the Declaration of Helsinki and the Nuffield Council.

Further, many of the reviews of ethical issues in clinical research that do exist focus on research work from one country and tend not to be representative of international perspectives. This lacuna prevents comparisons of laws and regulations as well as accepted norms of practice across countries. This has inhibited the identification of areas of agreement and disagreement, and led to claims that upon closer examination seem untenable.

In large measure, this textbook is an effort to remedy these and related lapses. One of its primary aims is to create a comprehensive and systematic synthesis of all the literature relevant to the ethics of clinical research, broadly construed. The textbook addresses all the topics of the ethics of clinical research and attempts to consider all the material written on each topic. It is comprehensive in covering the history of the triumphs of clinical

research as well as the scandals—and the formal guidance and scholarship that arose in the wake of these scandals. It analyzes such topics as the various perspectives of different groups of research participants, the assessment of risks and benefits, the operation of institutional review boards, informed consent, and what we know about conflicts of interest. We have tried to define the domain of the ethics of clinical research broadly, encompassing everything that might be relevant to the ethical consideration of a clinical research protocol, from regulations to the social context of research, from fraud and conflict of interest to confidentiality.

The survey is comprehensive in another way: It synthesizes both conceptual work, on issues such as payment to research participants or the involvement of women and children in research, and empirical studies on such topics as informed consent and financial conflicts of interest.

The survey is comprehensive in yet a third way: It focuses not just on the United States but on ethical debates, guidelines, regulations, studies from all over the world. Largely because so much clinical research has been funded by and conducted in the United States, much of the early development of research ethics occurred in the United States. However, with the ongoing and ever accelerating expansion of biomedical research in other countries—that is, with the globalization of research—there will be scandals, guidelines, regulations, and empirical studies from both developed and developing countries outside the United States. Doubtless this will affect research ethics. Although those in the United States often focus on individual rights, autonomy and informed consent, a shift to a global perspective may lead research ethics in new directions, perhaps with a greater emphasis on the impact of research on the wider community. In any case, to reflect and anticipate this globalization, we have tried to include authors from as many countries as possible.

The organization of the textbook is also meant to be instructive. Too often ethical considerations of controversial research trials are chaotic and haphazard, jumping from discussions of informed consent to considerations of risks and potential benefits to posttrial availability of interventions to use of stored tissue samples, and so on. This textbook aims to provide a systematic structure: Parts III through IX follow the logical sequence of development of clinical research protocols, and therefore of the ethical issues that have to be considered along the way. Research begins with defining the questions to be addressed, by what methodologies and in what relevant populations; only toward the end of preparing a research project are the issues of informed consent relevant. The organization and structure of the sections mirror this sequence to provide guidance through the research development process. Our intention is to have the formal structure of the book reemphasize how ethical issues can be addressed in a systematic manner, rather than in a disorganized manner. Hopefully, such an organized manner of analysis will contribute to improving the ethical evaluation of research protocols.

To achieve the goals of the textbook—comprehensiveness, systematic analysis, and wide ranging and international perspectives—we invited the world's leading authorities to contribute chapters. These individuals speak with tremendous knowledge and practical experience. We also sought to identify individuals from many different countries to ensure diverse perspectives and experiences. We believe the collection of over 85 authors from a variety of countries constitutes the very best scholarship on research ethics available.

Furthermore, we asked them to provide comprehensive summaries of their topics. After presenting the diverse perspectives fairly, we expected the authors to inject their own assessments. Why? Because this gives authors an opportunity to provide educated assessments of potential future developments of the ethical issues they have examined. We believe that this combination of comprehensive summaries and editorial perspectives of the author provides the best balance for interesting, informative, and vibrant chapters.

Our aim was to provide a book useful in training researchers and others. Over the last decade, there has been substantial emphasis on capacity development related to research in developing countries. Some of this attention has also focused on improving skills in the ethical review of research. Simultaneously, it has been observed that researchers and members of research review committees in developed countries have lacked training and knowledge in the ethical conduct of clinical research. Consequently, there has been recognition of the needs and efforts in both developed and developing countries to provide training in research ethics. Unfortunately, useful educational materials are lacking. This textbook is part of our efforts to address this deficiency. We hope teachers and students of research ethics will find it useful.

We are acutely aware that at best this textbook provides substantive guidance. This is necessary but not sufficient for ethical clinical research. In 1931, Germany enacted what many believe to be the first systematic, national research ethics guidelines. While these guidelines were in place, the Nazi violations were occurring. Guidelines are not self-enforcing. Researchers, research ethics committees, regulators, and others must enforce the rules. However, enforcers need to know what to enforce. A textbook such as this is absolutely necessary. Before we can enforce, we need to elucidate and specify what needs enforcement. We therefore hope that this is a book that researchers, members of ethics review committees, bioethicists, students, patient advocates, regulators, and others can consult to obtain clear guidance regarding the issues they confront. We hope it becomes a reliable and valued reference work.

In preparing any book, editorial decisions must be made that themselves will be controversial. How broadly or narrowly to define the textbook's subject? What specific topics to include or exclude? How much space to allocate to each topic? Which of several authorities in a particular area should be invited to author a chapter? The editors have wrestled with each of these issues and many others. There is no escaping contentious decisions that will annoy and even offend people. We accept full responsibility for these

choices, with the knowledge that this is a work of contemporary reflection and scholarship, not the final account of research ethics. As research proceeds and additional challenges arise, further constructive analyses and refinements will be necessary. This is for future editions.

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